Q1 $\boxed{C}$

$R = 15 + 35 + 25 = 75\,\Omega$

$10 + 40 = 50$

$75 \parallel 50 = \frac{75 \times 90}{75 + 50} = 30$

$P = \frac{90^2}{3} = 270\,W$

$\boxed{Q2}$ $(18 + 12) \parallel 60$

$\frac{30 \times 60}{30 + 60} = 20\,\Omega$

$50 \parallel 70$

$\frac{50 \times 75}{50 + 75} = 30$

$30 + 20 \parallel 50$

$\frac{50^2}{2 \times 50} = 25$

$P = (80 \times 10^{-3})^2 \times 20$

$= 128\,W$

$\frac{20 \times 60}{20 + 60} = 15\,\Omega$

$15 + 25 \parallel 40$

$\frac{40 \times 40}{40 + 40} = 20\,\Omega$

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$\boxed{Q_2}$

$3000 \parallel 1000$

$$\frac{3000 \times 1000}{3000+1000} = 750\,\Omega$$

$750 + 750 = 1500\,\Omega$

$3000 \parallel 2000$

$$\frac{3000 \times 2000}{3000+2000} = 1200\,\Omega$$

$1200 + 300 = 1500$

$1500 \parallel 1500$

$$\frac{1500}{2} = 750$$

$750 + 3000 + 250$

$R_{eq} = 4000$

$$P = \frac{400}{4000} = 0.1\,W$$

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$\boxed{Q_5}$

$6 // 7 + 5$

$$\frac{6 \times 12}{6 + 12} = 4\,\Omega$$

$$V(6) = \frac{4000}{800 + 4000} = 6V$$

$$V(5) = \frac{5000}{5000 + 7000} = 2{,}5V$$

Q6)

$$a = \frac{60 \times 15}{60 + 15} = 12\,\Omega$$

$$\frac{45 \times 30}{45 + 30} = 18$$

$$150 // 50 = 25\,\Omega$$

$$V(25) = 12{,}5$$

$$b) = \frac{12{,}5}{25} = \frac{1}{2}\,A$$

$$c)\ I(50) = \frac{50 // 150}{50} = 25$$

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